

Cybersecurity Incident Report: Suspicious DNS Traffic

Environment: Virtual Lab

Tools Used: Wireshark, dnscat2

1. Summary of the Incident

During routine network traffic monitoring, I observed an unusual pattern of DNS requests originating from a single device. These requests were being sent to an unknown domain at an abnormally high frequency. Upon further inspection, I suspected that DNS tunneling was being used to exfiltrate data from the network stealthily. This type of attack allows data to be transferred using DNS queries, bypassing traditional security measures.

2. Analysis and Cause of the Incident

Key Findings from Traffic Logs:

- Unusual DNS Queries: The device repeatedly sent requests to malicious.example.com.
- Encoded Payloads: The requests contained seemingly random strings, potentially representing base64-encoded data.
- High Traffic Volume: The number of DNS requests was significantly higher than normal, raising suspicion of automated activity.

Possible Cause:

The suspicious activity suggests that malware on the affected device was communicating with an external Command and Control (C2) server.

This could have been caused by:

1. A phishing attack that led to malware installation.
2. An insider threat attempting to bypass network security.
3. Misconfigured DNS security settings that allowed unauthorized external queries.

3. Incident Response and Mitigation

Immediate Actions Taken:

- Captured Network Traffic: Logged and analyzed packets using Wireshark.
- Blocked the Malicious Domain: Configured firewall rules to prevent further communication.
- Isolated the Affected Device: Removed it from the network to prevent further spread.
- Performed a Malware Scan: Checked for malicious software on the compromised endpoint.
- Enabled DNS Logging: Set up monitoring to track future suspicious activity.

Prevention Strategies:

- Restrict DNS Traffic: Allow DNS queries only through approved internal resolvers.
- Deploy Network Intrusion Detection Systems (NIDS): Tools like Suricata or Zeek can identify DNS tunneling.
- Enhance Employee Security Awareness: Conduct training on phishing and malware risks.
- Regular Patch and Security Updates: Ensure all endpoints and DNS servers are updated to mitigate vulnerabilities.

4. Conclusion & Lessons Learned

This incident highlights the importance of continuous network monitoring and proactive threat detection. Through hands-on analysis, I learned how attackers leverage DNS tunneling to exfiltrate data undetected and how to effectively detect, analyze, and mitigate such threats.